SCOTECI: Synthetic Chain-of-Thought Demonstrations for Document-Level Event Causality Identification

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Abstract

Large language models (LLMs) applied to Event Causality Identification (ECI) exhibit *causal hallucination*: near-total recall paired with unusable precision (Gao et al., 2023). Prior LLM-based scaffolds for ECI, such as Dr.ECI and SERE, fight this at the *pair* level, issuing one call per candidate event pair with bare input-to-label demonstrations (Cai et al., 2025; Hao et al., 2026). We propose SCOTECI, which instead (i) reformulates LLM-based ECI at the *document* level, labelling every pair of a document jointly in a single call under a structured reasoning protocol with explicit coherence rules, and (ii) equips its few-shot demonstrations with *synthetic chain-of-thought*: gold-guided reasoning generated by an LLM and then decontaminated so it reads as genuine discovery rather than answer confirmation. The method needs no fine-tuning, no human-written rationales, and no external knowledge base. On Causal-TimeBank, SCOTECI reaches F1 48.1 (precision 61.1) against F1 20.0 for the best prior LLM method — a +28.1 F1 gain obtained by breaking out of the degenerate high-recall/low-precision regime rather than trading recall for precision, on the dataset where positives are sparsest and hallucination is costliest. It also attains the best precision on MAVEN-ERE (40.8) and matches SERE's F1 on EventStoryLine (51.7 vs. 49.9). Beyond the headline numbers, we present an evidence-driven analysis of the method itself: majority-vote aggregation over demonstration ensembles is empirically F1-optimal, precision *rises* rather than falls with document causal density, and synthetic chain-of-thought demonstrations help chat-tuned backbones but measurably hurt a reasoning-tuned one.

1 Introduction

Event Causality Identification (ECI) asks, given a document and its marked event mentions, whether a directed causal relation holds between each pair of events (Cheng et al., 2025). It is a building block for event forecasting, causal question answering, and narrative and discourse understanding, and has been pursued along two largely disjoint paths: fine-tuned encoders and, more recently, prompted LLMs. Fine-tuned encoders — BERT/RoBERTa-era models augmented with event-relational graphs or external knowledge — are strong in-domain but require per-corpus annotation and generalize poorly outside the corpus they were trained on (Cai et al., 2025; Hao et al., 2026). Prompted LLMs sidestep the annotation cost entirely, but exhibit *causal hallucination*: on Causal-TimeBank, a plain zero-shot or chain-of-thought prompt reaches 80–96% recall with precision below 10% (Table 1, Base/CoT rows) — the model asserts a causal link almost anywhere a temporal or narrative connection exists (Gao et al., 2023).

The LLM scaffolds designed to fix this — Dr.ECI's four-agent discovery/reasoning pipeline (Cai et al., 2025), SERE's structurally retrieved demonstrations over ConceptNet paths and syntax-tree edit distance (Hao et al., 2026), and DICP's learned deep in-context prompts (Mu et al., 2025) — all operate at the *pair* level: one LLM call per candidate event pair, re-paying the full document context every time. This has three consequences. First, cost scales as $O(|\text{pairs}|)$ calls per document ($O(n^2)$ in the number of mentions), each of which resends the whole document. Second, each pair is judged in isolation, so nothing enforces symmetry or transitivity across the pairs of the same document, and the model never sees the document's causal structure as a whole. Third, these scaffolds were engineered around the specific weaknesses of 2023–2024-era models; as base models improve, the scaffold increasingly caps rather than unlocks their capability.

We propose SCOTECI, built on two ideas that invert this design. The first is to operate at the *document* level: a single call labels every pair of a document jointly, guided by an in-prompt reasoning protocol (mention pre-filter → per-pair arguments for and against → plausibility ranking → final JSON) together with explicit coherence rules — symmetry and dataset-specific transitivity — that are checkable only because the whole graph is produced at once. This shifts the burden of consistency from external orchestration to the model's own long-context reasoning, which is a deliberate bet: the method should get better, not merely cheaper, as the underlying model improves.

The second idea addresses what document-level demonstrations should look like. Bare input→gold-JSON demonstrations teach the model the output format but not how to reason about it, and document-level human-written rationales for ECI do not exist and would be expensive to produce. Since gold labels for training documents *are* available, we generate the reasoning instead: a demonstration document is shown to an LLM together with its own task prompt and its gold labels, and the model is asked to write reasoning that leads naturally to those labels (*gold-guided generation*); the response is then rewritten from scratch to remove any phrasing that betrays foreknowledge of the answer (*decontamination*), so that every sentence reads as discovery rather than confirmation. An optional three-step variant inserts a blind attempt before the gold-guided pass, preserving more of the texture of a genuine, imperfect first try. Each demonstration becomes a (user, assistant) exchange in the prompt, so the model effectively "remembers having solved" a handful of documents correctly, reasoning included.

On Causal-TimeBank, SCOTECI reaches F1 48.1 (precision 61.1, recall 39.6) against F1 20.0 for the best prior LLM method (SERE) — a +28.1 F1 gain, with precision more than four times any LLM baseline in the comparison (Table 1). This is the dataset where positives are sparsest and hallucination is most punishing, and it is also the dataset where the balanced precision/recall regime most clearly replaces the degenerate high-recall/near-zero-precision one that every baseline exhibits. SCOTECI additionally attains the best precision on MAVEN-ERE (40.8) and an F1 that edges out SERE on EventStoryLine (51.7 vs. 49.9). All three SCOTECI numbers are reported on the intra-sentence subset of pairs, which is the only subset SERE itself evaluates, so this is the fair like-for-like comparison; full-document (intra- and inter-sentence) prediction is a capability the document-level formulation affords — one call covers all pairs regardless of how many of them cross a sentence boundary, whereas a pair-level method pays for every additional inter-sentence pair as another call — but is not yet measured for SCOTECI and is left to future work (see Limitations).

Beyond the headline table, we use SCOTECI's logged predictions to answer questions the original design left open. We show that the majority-vote aggregation used across resampled passes is empirically F1-optimal rather than merely "precision-preserving" as assumed, that precision *rises* rather than falls with a document's causal density — the opposite of the attention-dilution failure mode we originally expected — and that synthetic chain-of-thought demonstrations help chat-tuned backbones but measurably hurt a reasoning-tuned one, a finding with direct implications for which future backbones the method should target.

Our contributions are:

- A document-level reformulation of LLM-based ECI, with a structured in-prompt reasoning protocol and explicit coherence rules, that issues $O(1)$ LLM calls per document instead of $O(|\text{pairs}|)$ and produces predictions that are consistent across a document's pairs by construction.
- A synthetic chain-of-thought demonstration pipeline — gold-guided generation followed by decontamination — that is model-agnostic and needs no human-written rationales or external knowledge base.
- State-of-the-art LLM results on Causal-TimeBank by a wide margin, with balanced precision/recall on all three evaluated datasets, plus an evidence-based analysis of aggregation, document density, backbone interaction, cost, and robustness that goes beyond the headline comparison.

2 Related Work

2.1 Supervised and Knowledge-Enhanced ECI

Before LLM prompting, ECI was dominated by fine-tuned encoders that inject structural or external signal into a BERT/RoBERTa-style backbone: event-relational graph transformers that propagate information across a document's mention graph (Chen et al., 2022), sparse event representations for discriminative document-level event-relation extraction (Yuan et al., 2023), joint identify-while-learn objectives that couple mention and relation supervision (Liu et al., 2024), knowledge-guided binary question-answering formulations of the pairwise decision (Wang et al., 2024b), and heterogeneous graph contrastive transfer for zero-shot cross-lingual ECI (He et al., 2024). These remain the strongest in-domain systems on their respective benchmarks, but each requires gold-annotated training data for the specific corpus (often the specific label scheme) it is evaluated on, and transfer across corpora or languages is itself an active research problem rather than a given (He et al., 2024). SCOTECI requires no task-specific training at all: the training split is used only as a pool of few-shot demonstrations.

2.2 LLM-based ECI

Gao et al. (2023) first showed systematically that ChatGPT-class models over-predict causal links — causal hallucination — and established the preprocessing and evaluation protocol that we and our baselines follow. Two scaffolds subsequently targeted this failure mode at the pair level. Dr.ECI decomposes each candidate pair into a discovery stage (a Causal Explorer and a Mediator Detector) and a reasoning stage (Direct and Indirect Reasoners), guided by causal-pattern DAG priors (Cai et al., 2025). SERE retrieves structurally similar pair-level examples using ConceptNet shortest-path edit distance, dependency-tree edit distance, and causal-pattern filtering, and balances positive and negative demonstrations by construction (Hao et al., 2026). Closer to our synthetic-rationale idea, Wang et al. (2024a) frame ECI through a synthetic-control, counterfactual-estimation lens, and DICP learns deep in-context prompts per pair rather than retrieving them (Mu et al., 2025). All four of these methods (i) prompt once per pair and (ii) show label-only demonstrations.

Two very recent methods are closer still to specific pieces of our design. Like SCOTECI, CogECI operates at the document level and uses an LLM to produce reasoning-adjacent signal — grounded context pseudo-labels extracted from the document — but consumes that signal as supervision for a separately trained graph network (a Sentence Graph Module and a Dual-Stream Causality Reasoning Module) rather than as an in-context demonstration, so it remains a fine-tuned method with an LLM-assisted labelling step, not a training-free prompting method (Zhang et al., 2026). Zhao and Yan (2026) generate chain-of-thought traces specifically to reduce causal hallucination in ECI, the same failure mode we target, but their traces are training data — used to fine-tune small ($\leq$1.5B-parameter) models against a dedicated causal-hallucination-rate metric — rather than in-context demonstrations for a frozen, larger model, and their unit of generation is a single relation rather than a whole document. SCOTECI differs from all of the above on both axes at once: it prompts once per document rather than once per pair or per relation, it shows demonstrations that carry explicit reasoning rather than labels or pseudo-labels, and it requires no fine-tuning whatsoever.

2.3 In-Context Learning and Synthetic Rationales

Chain-of-thought prompting improves multi-step reasoning by asking a model to externalize intermediate steps before its final answer (Wei et al., 2022), building on the broader observation that large models are effective few-shot learners given the right demonstrations (Brown et al., 2020). In-context learning, however, is highly sensitive to which demonstrations are shown and how they are phrased (Liu et al., 2022; Min et al., 2022). Where rationales are not naturally available, prior work has synthesized them: Auto-CoT clusters queries and applies zero-shot CoT to generate rationales for demonstration selection (Zhang et al., 2023); STaR bootstraps rationales through *rationalization* — generating reasoning conditioned on a known answer — but consumes the result as fine-tuning data (Zelikman et al., 2022); step-by-step distillation similarly extracts rationales as a training signal for smaller student models (Hsieh et al., 2023). SCOTECI uses the same rationalization trick as STaR, but differently: the synthesized reasoning is consumed *at inference time* as in-context demonstrations rather than as fine-tuning data, it operates at document rather than sentence/example scale, and it adds an explicit decontamination pass to strip answer-leakage phrasing — without which, we find, demonstrations teach the model to assert conclusions rather than to reason toward them (§6.1).

3 Task Formulation

Given a document D with event mentions marked inline in `<ID word>` format, $M = \{m_1, \ldots, m_n\}$, and a candidate pair set P —

the annotated pair list for corpora that provide one (the *constrained* setting, used for MECI and Causal-TimeBank) or all unordered mention pairs otherwise (the *open-ended* setting, used for MAVEN-ERE and EventStoryLine) — the task is to produce a complete labelling $\ell : P \to \{causes, causedby, norel\}$: a unified, directed three-way label space into which every corpus' native annotation scheme is mapped before evaluation. Concretely: MECI's CAUSE-EFFECT/EFFECTCAUSE/NOREL labels map one-to-one onto *causes*/*causedby*/*norel*; MAVEN-ERE's CAUSE and PRECONDITION relation types are both collapsed onto the same directed causes/causedby pair; EventStoryLine's PRECONDITION and FALLING_ACTION relations are collapsed the same way; and Causal-TimeBank's CLINK annotations map directly, restricted to the intra-sentence subset (286 of 318 CLINKs, 90%; the remaining cross-sentence CLINKs are dropped at preprocessing, so Causal-TimeBank has no genuine inter-sentence positives and consequently no intra/inter split anywhere in this paper). For MAVEN-ERE, EventStoryLine, and Causal-TimeBank, *norel* pairs are not separately annotated in the source data; they are the complement of the positive pairs within P.

Document-level inference means the model produces ℓ for *all* of P in a single generation, formatted as a JSON array, in contrast to pair-level inference, which issues $|P|$ independent queries, one per candidate pair.

We follow Gao et al. (2023)'s preprocessing and evaluation protocol for comparability with Hao et al. (2026) and Cai et al. (2025): micro precision/recall/F1 over the positive (causal) classes, *norel* excluded from the denominator, reported on the intra-sentence subset where a corpus distinguishes intra- from inter-sentence pairs — the subset SERE itself evaluates, and hence the fair comparison point. We additionally adopt a binary framing throughout: once a pair is identified as causal, whether it is scored as *causes* or *causedby* is collapsed into a single positive class before computing metrics, since the object of interest in this work is *whether* a causal link exists, not which of the two mentions is cause versus effect. Directed (non-collapsed) metrics are logged for every run but are not the primary quantity we report or optimize for.

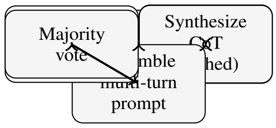

Figure 1: Overview of SCOTECI: neighbour retrieval (§4.3), synthetic CoT generation (§4.4), multi-turn prompt assembly and a single document-level call (§4.2), and majority-vote aggregation across N resampled passes (§4.5).

4 Method

4.1 Overview

The pipeline (Figure 1) proceeds in four stages for each test document: retrieve $k = 3$ neighbour training documents (§4.3); generate and cache a synthetic chain-of-thought demonstration for each neighbour (§4.4); assemble a multi-turn prompt in which every demonstration appears as a past (user, assistant) exchange, followed by the test document; and issue a single document-level call, under the structured reasoning protocol of §4.2, whose parsed JSON output is optionally aggregated across N resampled passes (§4.5). Two components carry the paper's contribution — the document-level structured prompt (§4.2) and the synthetic chain-of-thought demonstrations (§4.4); neighbour retrieval (§4.3) and aggregation (§4.5) are deliberately kept simple, and §6.1 measures what each of the four pieces contributes.

4.2 Document-Level Structured Prompting

One call handles the entire document: every pair in P is requested at once, and the model sees the full narrative rather than a windowed snippet around a single pair. The system prompt enforces an in-prompt reasoning protocol with four ordered steps. First, a mention-level pre-filter excludes mentions only via an affirmatively argued reason (e.g., "m is a location with no agentive role"), never a blanket dismissal, and lists which mentions survive. Second, for every retained pair the model lays out arguments *for* a causal link (an explicit connective, a plausible mechanism, counterfactual asymmetry, supporting discourse structure) and arguments *against* it (a merely temporal relation, a confounding third event, a reporting verb, excessive distance). Third, the surviving pairs are ranked qualitatively by causal plausibility, without committing to labels yet. Fourth, the model emits a final JSON labelling of *every* requested pair; anything excluded in step one defaults to *norel*.

A precision-first rule block targets causal hallucination directly: a two-signal requirement for non-overt links (at least two independent pieces of textual evidence, not one), a stricter evidentiary gate for inter-sentential or otherwise long-range pairs, a default to *norel* under genuine uncertainty, an explicit confounder/proximal-cause check, a rule that reporting and purely temporal connectives are non-causal on their own, and a counterfactual direction test. One dataset-specific instance of this rule block matters enough to disclose explicitly: on Causal-TimeBank, the active prompt additionally requires an *explicit causal marker* directly connecting the two events for a *causes/ causedby* label, defaulting to *norel* when no such marker is present. Causal-TimeBank's CLINK annotation is itself markedly explicit-connective (roughly half of gold links carry an overt causal marker), so the rule is dataset-appropriate rather than a silent addition to "the protocol"; a matched ablation on the reported backbone (§6.1) measures its isolated contribution at +10 to +19 F1 out of the +28.1 F1 headline gain over SERE, and we state the rule here rather than folding it into the generic rule block above.

Finally, the prompt states two coherence rules that are checkable only because the whole pair set is produced jointly: symmetry (*causes*$(i, j) \Leftrightarrow$ *causedby*(j, i)) and dataset-specific transitivity rules (e.g. chains of MAVEN-ERE's PRECONDITION relation compose transitively; full rule sets are provided with the released configuration). In the runs behind Table 1 these rules are stated in-prompt only — the pipeline also supports a post-hoc tool that could mechanically check or prune violations after generation, but that tool was disabled (`enable_tools=False`) throughout, so their effect here is whatever the model achieves by reading the rules, not by external enforcement; §6.1 discusses why we cannot yet isolate their contribution from the rest of the protocol.

This design leverages better models on purpose: the protocol asks for exactly the kind of global, long-context reasoning that frontier models are differentially good at, whereas pair-level scaffolds cap that ability at the width of a single pair.

4.3 Neighbour-Document Retrieval

The demonstration pool is the training split of the same corpus, capped to 100 documents (`pool_size`) to bound the cost of synthetic CoT generation (§4.4): every demonstration document needs its reasoning generated once, so a larger pool means more one-off generation cost, amortized by caching across the run but not free. Table 1's SCOTECI row uses *random* selection: $k = 3$ demonstrations sampled uniformly from the pool for each call. We also implement three retrieval-based alternatives: TF-IDF cosine similarity over the full document text, sentence-embedding cosine similarity, and Jaccard similarity over the set of event-mention strings; the first two are compared against random selection in §6.1, while mention-set Jaccard is implemented but has not yet been evaluated in any run. Every demonstration is a *whole document with its complete gold labelling*, including its *norel* pairs, so each demonstration carries explicit negative evidence for free — the pair-level analogue of which SERE must engineer by explicitly balancing positive and negative retrieved examples.

4.4 Synthetic Chain-of-Thought Generation

Label-only demonstrations teach the output schema but not the reasoning behind it; document-level human rationales are unavailable and would be prohibitively expensive to write; but gold labels for training documents *are* available, so rationales can be synthesized instead. In the default 2-step protocol, each demonstration document is (i) shown under the exact task prompt together with its gold labels, with an instruction to write reasoning that "leads naturally to these labels from the text alone" without revealing foreknowledge (*gold-guided generation*), and then (ii) rewritten from scratch to purge leakage phrasing ("the labels show…", "as indicated by the expected output…"), so every sentence reads as analysis rather than confirmation (*decontamination*). A 3-step variant inserts a blind attempt with no gold access before the gold-guided correction step, so the sequence becomes blind attempt $\rightarrow$ gold-based correction (fixing wrong conclusions while keeping correct reasoning) $\rightarrow$ decontamination; this preserves more of the texture and difficulty profile of a genuine, imperfect first try. Table 1's SCOTECI row uses the 3-step variant; a limited 2- vs. 3-step comparison exists (§6.1), but the difference falls within run-to-run noise on the runs available so far.

Each demonstration becomes a (user task turn, assistant CoT turn) exchange preceding the test document, so the model effectively "remembers having done" k documents correctly, reasoning included; this is compatible with multi-step task prompts, which receive one synthetic response per step. Demonstrations are generated once per (document, protocol) and cached for the remainder of the run; unless noted otherwise, the generator model is the same as the inference model (self-generated rationales) — whether a stronger generator can lift a weaker reasoner is a question we leave open (see Limitations). Decontamination is not currently an optional step: both the 2- and 3-step protocols apply it unconditionally, so while we motivate it as necessary (§2), we do not yet have a controlled ablation that isolates its effect from a from-scratch demonstration generated without it; §6.1 reports what the current logs can and cannot support on this point.

The relation to STaR-style rationalization (Zelikman et al., 2022) is direct: the same trick — condition on the answer to obtain reasoning — but used as in-context demonstrations at inference time rather than as fine-tuning data, and with an explicit decontamination pass that we motivate as necessary for the demonstrations to teach reasoning rather than assertion.

4.5 Inference and Aggregation

Decoding uses temperature 0 with strict JSON parsing and bounded retries; a document that exhausts all retries is scored as entirely *norel* rather than excluded from evaluation (2 of 183 Causal-TimeBank documents fall into this category for the headline run; EventStoryLine and MAVEN-ERE have none). Pairs absent from a parsed response also default to *norel*.

Table 1's SCOTECI row uses $N = 3$ resampling passes per document with a per-pair majority vote, ties broken to *norel*; each pass draws its own set of k demonstrations independently (`distinct_fewshots=True`), split round-robin from a larger draw so the three passes' demonstration sets are disjoint. We originally intended pair-order reshuffling across passes to be the source of inter-pass disagreement, but at temperature 0, and with the candidate pair list identical across passes for the constrained-pair-list datasets, reshuffling has no effect once the pair list itself does not change. What actually differs between passes is *which* demonstrations they see. "Resampling with majority vote" is therefore better understood as an ensemble over demonstration sets, plus whatever provider-level nondeterminism survives temperature 0, rather than as stochastic resampling of an otherwise-fixed call. §6 shows that this reframing explains why passes disagree at all — false positives are pass-specific while true positives are stable across the ensemble — and that the resulting majority-vote rule is empirically F1-optimal on all three datasets, not merely precision-preserving by assumption.

Cost per document is 1 call under SCOTECI (N under resampling), plus the amortized, cached cost of generating each demonstration's synthetic CoT once (2 or 3 extra calls per unique demonstration document, shared across every test document that retrieves it). This contrasts with $|P|$ calls per document for pair-level methods, each resending the document, plus SERE's per-pair retrieval infrastructure (ConceptNet and a dependency parser) and Dr.ECI's up to four agent calls per pair; §6.3 quantifies the resulting gap using the tokens and dollar costs logged for the headline runs.

5 Experiments

5.1 Datasets and Metrics

We evaluate on three English ECI corpora spanning news and Wikipedia text, chosen to vary substantially in positive-pair density and scale.

EventStoryLine (ESC; Caselli and Vossen, 2017), version 0.9: 258 news documents across 22 topics, with causal (PRECONDITION/ FALLING_ACTION) links annotated over ECB+ event chains. Following the corpus's original convention, documents in topics 37 and 41 are held out as a development split; we evaluate on the remaining 233 documents (the `train` split), with 5-fold separation ensuring a document's few-shot demonstrations never come from its own fold.

Causal-TimeBank (CTB; Mirza et al., 2014): 183 news documents, 318 CLINK-annotated causal pairs, of which 286 (90%) are intra-sentence (§3) — the sparsest-positive of the three corpora and, consequently, the one where causal hallucination is most costly. As with ESC, we evaluate on the full corpus, with 10-fold demonstration separation.

MAVEN-ERE (Wang et al., 2022): a large-scale Wikipedia corpus with 57,992 causal pairs (CAUSE and PRECONDITION, collapsed as in §3) across its full release. We evaluate on its 710-document test split (33,144 candidate pairs), using the training split as the demonstration pool.

All preprocessing and label mapping follows Gao et al. (2023)'s protocol (§3) for comparability with our baselines. We additionally have exploratory results on MECI (Lai et al., 2022), a multilingual ECI corpus the pipeline fully supports;

those results are noisy across repeated runs of the same configuration and are not yet at the standard of the three corpora above, so we omit them from the main comparison (see Limitations).

5.2 Baselines

We compare against the LLM-prompted methods surveyed in §2: **Base**, a task description with no reasoning scaffold; **CoT** (Wei et al., 2022), step-by-step reasoning before the final answer; **Dr.ECI** (Cai et al., 2025); and **SERE** (Hao et al., 2026). All four are pair-level: one call per candidate pair. The GPT-4o-mini and Gemini-1.5-pro rows are reproduced from Hao et al. (2026)'s Table 2; the PaLM2 Dr.ECI row is reproduced from Cai et al. (2025). We do not re-run any baseline ourselves, so differences between SCOTECI and these rows reflect both the document-level reformulation and a different backbone; we do not yet have a same-backbone pair-level control that would separate the two effects (§6.3 discusses why, and what evidence we have instead).

5.3 Implementation Details

Unless noted otherwise, SCOTECI uses `deepseek/deepseek-v4-pro` (via OpenRouter) as both the inference and the CoT-generation backbone, at temperature 0. This is a mid-tier open model rather than a frontier one; §6.4 reports what a limited, single-seed comparison across backbones (including a frontier model) suggests about how the method scales with model quality, and §6.3 shows it is already competitive with a frontier backbone on a cost-per-F1 basis. Demonstrations use $k = 3$, random selection from a pool of 100 training documents, and the 3-step synthetic CoT protocol (§4.4). The headline numbers use $N = 3$ resampling passes with distinct demonstration sets per pass and a majority vote (§4.5). Exact prompts (per dataset), CoT-generation prompts, and coherence rule sets are versioned with the released configuration.

Synthetic CoT generation costs 2 (2-step) or 3 (3-step) additional LLM calls per unique demonstration document, amortized across every test document that draws it from the shared pool. This cost is not separately metered in our logs but is included in the total costs reported in §5.4 and §6.3; isolating it precisely would require either per-call cost tagging or mining token counts from the cached CoT text, which we leave for future tooling.

5.4 Main Results

Table 1 reports precision, recall, and F1 for SCOTECI against the LLM baselines surveyed in §5, all evaluated on the intra-sentence subset (§3).

Causal-TimeBank (the headline). SCOTECI reaches F1 48.1 against 20.0 for the best LLM baseline (SERE) — a +28.1 F1 gain — with precision 61.1 where no LLM baseline exceeds 13.8. This is the dataset with the sparsest positives and the costliest hallucination, and it is where the degenerate high-recall/ low-precision regime is broken most clearly: every baseline sits at recall ≥79.6 with precision ≤13.8, while SCOTECI inverts the ratio entirely (P 61.1, R 39.6). A matched ablation (§6.1) measures roughly +10 to +19 F1 of this gain as coming from a dataset-specific prompt rule rather than from the document-level scaffold alone (§4.2); we report that decomposition in full there rather than overstating what the scaffold alone contributes here.

MAVEN-ERE. SCOTECI attains the best precision in the table (40.8) and an F1 of 48.4, exceeding SERE's 42.3 by 6.1 F1, on the full 710-document test split (33,144 candidate pairs) — making this the most heavily evaluated of the three columns.

EventStoryLine. SCOTECI reaches F1 51.7 (P 40.7, R 70.9), edging out SERE's 49.9 while trading some of SERE's precision for recall. Unlike CTB and MAVEN-ERE, SCOTECI's precision on ESC (40.7) sits closer to the CoT/Dr.ECI baseline range than to SERE's, so the gain here comes primarily from recall rather than precision — the opposite pattern from CTB.

Reading across rows. Base, CoT, and Dr.ECI all sit at recall ≥75 with precision ≤31 on every dataset, the hallucination signature described in §1. SERE trades recall for precision but never exceeds P 48.3. SCOTECI attains the best precision on two of three datasets (CTB, MAVEN-ERE) and the best F1 on two of three (ESC, MAVEN-ERE), without ever collapsing into the degenerate low-precision regime that Base/CoT/Dr.ECI exhibit everywhere.

On significance. Each cell in SCOTECI's row is a single configuration's result (with $N = 3$ resampling folded in, §4.5), not a multi-seed mean. Repeated identical-configuration runs elsewhere in our logs put the run-to-run standard deviation on 50-document subsets at roughly 2–4 F1 (ten repeats of

Method	ESC P	ESC R	ESC F1	CTB P	CTB R	CTB F1	MAVEN-ERE P	MAVEN-ERE R	MAVEN-ERE F1
PaLM2:									
Dr.ECI[†]	29.0	75.4	41.9	-	-	13.0	22.0	80.0	34.5
GPT-4o-mini:									
Base	28.4	82.9	42.3	5.4	80.5	10.1	23.1	91.5	36.9
CoT	29.4	80.6	43.1	6.3	79.6	11.6	25.4	87.9	39.4
Dr.ECI	31.0	**90.2**	46.1	8.2	91.2	15.1	26.0	93.0	40.7
SERE	**48.3**	51.5	**49.9**	13.8	36.3	20.0	34.5	54.6	**42.3**
Gemini-1.5-pro:									
Base	24.0	80.8	37.0	4.7	**95.6**	9.0	21.2	**95.1**	34.6
CoT	25.1	85.4	38.7	4.7	88.5	8.9	23.3	89.2	37.0
Dr.ECI	27.6	82.0	41.3	7.2	88.5	13.3	23.4	91.7	37.3
SERE	36.5	59.3	45.2	9.9	69.9	17.4	29.9	60.2	39.9
Ours:									
SCOTECI*	40.7	70.9	51.7	**61.1**	39.6	**48.1**	**40.8**	59.6	48.4

Table 1: Main results. Baseline rows reproduced from Hao et al. (2026) (their Table 2). Best score per column is in bold across all rows. †: results from Cai et al. (2025). CTB has no intra/inter split, so its column is directly comparable throughout. On ESC and MAVEN-ERE, SERE itself evaluates intra-sentence pairs only, so SCOTECI's intra-sentence scores (*) are the fair like-for-like comparison; full-document (intra+inter) SCOTECI numbers are additional future work (see Limitations), not a gap in this comparison.

one Causal-TimeBank configuration ranged 39.8–51.0 F1), and a broader sweep of 31 byte-identical-configuration reruns has a median F1 spread of 0.02 and a 75th-percentile spread of 0.09 (0–1 scale). The Causal-TimeBank gain over SERE (+28.1 F1) and the MAVEN-ERE gain (+6.1 F1) are both well outside this noise band; the EventStoryLine gain (+1.8 F1) is not, and should be read as "at least competitive with," not "reliably better than," SERE on that dataset. §6.1 discusses what multi-seed reporting of the ablation deltas would require.

6 Analysis

6.1 Ablations

Every ablation below is reported as a prose delta against a matched configuration rather than as a new table, and calibrated against the noise floor established in §5.4: on 50-document subsets, run-to-run F1 standard deviation is roughly 2–4, and a broader sweep of byte-identical reruns puts the median spread at 0.02 and the 75th percentile at 0.09 (0–1 scale). Much of what is available today is single-seed, on an earlier backbone, or otherwise confounded, and we report that plainly rather than presenting underpowered comparisons as settled.

Explicit-marker rule (Causal-TimeBank), on the final backbone. Unlike the ablations below, this one is a clean, matched, current-backbone comparison rather than an inherited estimate: two matched pairs on `deepseek-v4-pro` (Causal-TimeBank, n=50, k=3 random demonstrations, pool 20, 3-step CoT) — one pair with resampling off, one with N=3 resampling — compare the explicit-marker prompt (§4.2) against an otherwise-identical prompt with the marker requirement removed. Without resampling: marker F1 39.6 vs. no-marker 21.0, a +18.6 F1 gain. With N=3 resampling: marker F1 47.5 vs. no-marker 37.2, a +10.3 F1 gain. Both deltas are well outside the noise floor and confirm, on the actual reported backbone, what the earlier v3.2-era decomposition only suggested: the marker rule does substantial, real work, though on `deepseek-v4-pro` its isolated contribution (+10 to +19 F1) is smaller than the roughly +30 F1 estimated on the older backbone — consistent with a stronger backbone needing the explicit hint less. This closes the gap the ablation audit behind this section originally flagged as missing.

Synthetic CoT vs. label-only demonstrations. On the Causal-TimeBank explicit-marker prompt, matched runs on an earlier backbone (`deepseek-v3.2`) put CoT demonstrations at F1 43.1 (mean of 13 runs) against 34.1 for label-only demonstrations (a single run), a +9.0 F1 gain. We do not yet have a matched comparison on the final `deepseek-v4-pro` configuration, and §6.5 reports a complicating finding: the same substitution *hurts* a reasoning-tuned backbone by a comparable margin, so whether CoT demonstrations help is backbone-dependent rather than uniformly true.

Few-shot on/off and protocol on/off. An early-backbone Causal-TimeBank run with few-shot disabled entirely scored F1 40.9, *above* the label-only-demo condition above (34.1) on a single seed each — suggestive that badly-chosen demonstrations can cost more than they give, but not a clean result. Removing the structured protocol (the `*_norule*` prompt variants, demonstrations kept) shows small, noise-level differences on EventStoryLine (41.4 vs. 40.8–44.4) and MAVEN-ERE (33.6 vs. 34.8–36.6, earlier backbone); Causal-TimeBank's only no-protocol runs predate the explicit-marker rule and are not comparable. Neither comparison currently supports a quantitative claim about the structured protocol's contribution in isolation.

Demonstration selection. Random selection matches or exceeds TF-IDF and sentence-embedding retrieval at document scale: on MAVEN-ERE (matched setting, `deepseek-v4-pro`), random selection scores 45.0–47.1 F1 across four runs, embedding retrieval 37.5–46.6 across eight runs, and TF-IDF 39.6 on a single run. This echoes, at document scale, SERE's own report that naive retrieval can underperform random selection at pair level (Hao et al., 2026), and matches our choice of random selection for the headline table (§4.3). Mention-Jaccard selection is implemented but has not yet been run. A related trend: larger demonstration pools (more per-document diversity to sample from) trend positive, though not for free — diverse per-document demonstrations reduce prompt-cache reuse, and our headline runs read only 18–21% of input tokens from cache; a fixed demonstration set would be materially cheaper to serve but likely somewhat worse.

Number of demonstrations. All headline runs use $k = 3$. A single MAVEN-ERE comparison exists at $k = 6$ (39.4 vs. 38.4 for matched $k = 3$, both single runs); $k = 1$ and $k = 5$ have not been run. We cannot yet say whether document-level demonstrations saturate or degrade past $k = 3$, as pair-level demonstrations do past $k = 2$ in SERE.

2-step vs. 3-step CoT. Two matched pairs exist: EventStoryLine (2-step 41.8 vs. 3-step 40.6) and MAVEN-ERE, earlier backbone (2-step 38.9 vs. 3-step 39.3). Both differences are within run-to-run noise. We use the 3-step variant for the headline table on the strength of its intended benefit (preserving the texture of a genuine attempt, §4.4) rather than a measured advantage.

Decontamination on/off. We motivate decontamination as necessary (§4.4, §2), but the current implementation applies it unconditionally in both CoT protocols, so isolating its effect requires a configuration option that does not yet exist; a raw, non-decontaminated variant has been run only once, on a single document, as a smoke test. This ablation is missing, not merely underpowered, and is the clearest concrete item of future work this paper leaves (see Limitations).

Coherence rules on/off. The post-hoc coherence-checking tool was disabled in every run behind Table 1 (§4.2), so no run isolates the in-prompt coherence rules from the rest of the protocol; earlier tool-enabled runs use different prompts and models and are not comparable. This ablation, like decontamination, is currently missing rather than inconclusive.

6.2 Aggregation: Majority Vote Is Empirically F1-Optimal

§4.5 reframes resampling as an ensemble over demonstration sets rather than stochastic resampling. This lets us ask directly whether the shipped aggregation rule — majority vote of $N = 3$ passes, ties broken to *norel* — is actually the best choice, by sweeping the vote threshold $t \in \{1, 2, 3\}$ (predict causal iff at least t of the 3 passes agree) over the full per-pair predictions of the three headline runs, reconstructed from their raw per-pass traces.[1] Figure 2 plots precision, recall, and F1 at each threshold for all three datasets.

Majority voting ($t = 2$) is F1-optimal on all three datasets: union voting ($t = 1$) costs 3.5–5.2 F1 relative to it, and unanimous voting ($t = 3$) costs 0.8–3.0 F1. The reason is asymmetric reliability: on Causal-TimeBank, 70% of the false positives that union voting adds appear in only one of the three passes, while 72% of union's true positives survive into the majority — per-pass hallucinations are largely ephemeral, and the vote fil-

[1] Reconstructed rather than read from the logged per-pair predictions, which for these runs retain only gold $\cup$ final-majority pairs and would silently hide exactly the pass-specific false positives a union rule adds; the reconstruction reproduces each run's logged majority-vote metric within 0.1–1.1 F1, which we take as validating the reconstruction.

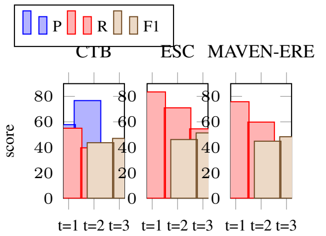

Figure 2: Precision, recall, and F1 as a function of the resampling vote threshold t, on the three headline runs. Majority (t=2) is F1-optimal everywhere.

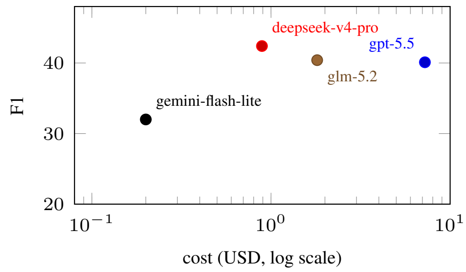

Figure 3: Cost-effectiveness: F1 vs. evaluation cost by backbone, MAVEN-ERE n=50 (single seed per point; deepseek-v4-pro and the gemini-flash-lite tier are plotted at the midpoint of their reported ranges).

ters them out, while genuine signal is stable across the demonstration ensemble. Unanimous voting is nonetheless a real, usable precision mode for applications that need it (Causal-TimeBank P 76.7, MAVEN-ERE P 49.0, EventStoryLine P 46.9), and union recall (55.0 / 75.8 / 83.5) measures the recall ceiling of the demonstration ensemble as a whole. Ties-to-*norel* (§4.5) is therefore not merely a reasonable default but the empirically correct one for our reported configuration.

6.3 Document-Level vs. Pair-Level, Same Backbone

The internal comparison that would most cleanly separate the document-level formulation's effect from the choice of backbone — the same model, forced into one-call-per-pair inference under the same rules — is not currently implemented in our pipeline (`prompts/esl_perpair.yaml` exists but was never run to completion as a controlled comparison), and we do not have it to report. What we do have is an arithmetic cost argument and a measured cost-effectiveness result.

On calls and tokens: a pair-level method issues $|P|$ calls per document, each re-sending the full document; on MAVEN-ERE's evaluated split alone, $|P| = 33{,}144$ against 710 single-call SCOTECI documents (plus $N = 3$ resampling and the amortized, cached cost of demonstration CoT generation). Dr.ECI additionally issues up to four agent calls per pair (Cai et al., 2025), and SERE's per-pair retrieval requires a ConceptNet lookup and a dependency parse for every candidate pair (Hao et al., 2026). Measured end-to-end costs for the three headline runs (inference plus resampling plus CoT generation) are \$7.49 (Causal-TimeBank), \$13.19 (EventStoryLine), and \$24.15 (MAVEN-ERE).

Separately, on the same MAVEN-ERE 50-document setting used for the model ladder (§6.4), our mid-tier backbone (`deepseek-v4-pro`) matches or exceeds a frontier model (`gpt-5.5`: F1 40.1 at \$7.23) at F1 39.6–45.2 for \$0.66–1.12 — roughly an order of magnitude better cost-per-F1 (Figure 3); `glm-5.2` reaches F1 40.4 at \$1.81 and `gpt-4o-mini` reaches F1 24.3. This is a stronger, already-measured version of the cost argument that a same-backbone pair-level control would have made: SCOTECI on a mid-tier open backbone is both more accurate and, per unit of F1, an order of magnitude cheaper than one plausible pair-level cost driver — using a frontier model to compensate for a weaker scaffold.

6.4 Scaling with Model Quality

The claim we set out to test is that the document-level formulation's advantage over pair-level prompting *widens* as the backbone improves, because it exposes rather than bypasses a model's long-context global reasoning. We do not yet have the evidence to make that claim in its strongest (crossover) form: doing so needs a pair-level arm at each point on a model ladder, and §6.3 explains why that arm does not exist yet. What we have instead is a single-seed, partially confounded ladder of SCOTECI scores across backbones on MAVEN-ERE (intra-sentence, $n = 50$): `gpt-5.5` 40.1, `glm-5.2` 40.4, `deepseek-v4-pro` 38.4–45.2 (varying with pool size and selection), `deepseek-v3.2` 37.8–39.9, `qwen3-30b-thinking` 35.9, `gemini-2.5-flash-lite` 34.0, `gemini-3.1-flash-lite` 30.1, and `gpt-4o-mini` 24.3 (Figure 4). The direction is consistent with "bigger and more capable backbones score higher," and an isolated Causal-TimeBank data point is suggestive in the expected direction — with few-shot disabled,

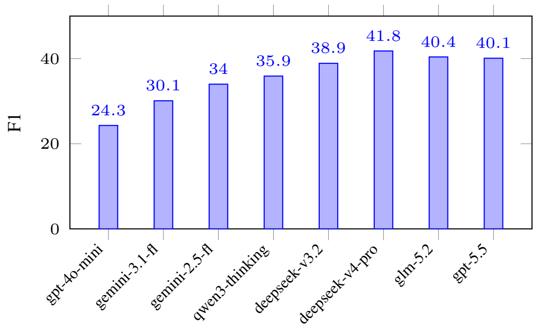

Figure 4: SCOTECI F1 across backbones, MAVEN-ERE intra, n=50, single seed per point (deepseek-v4-pro and deepseek-v3.2 shown at the midpoint of their reported ranges; x-axis ordered low-to-high by this measurement, not by a fixed independent tier ranking).

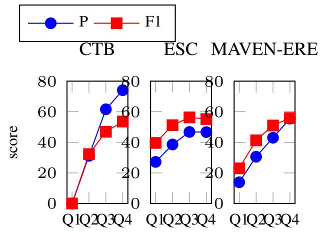

Figure 5: Precision and F1 by document causal-density quartile (Q1 sparsest – Q4 densest), pooled binary-intra, on the three headline runs.

`gpt-5.4-mini` collapses to 9.5 F1 where `deepseek-v3.2` reaches 35.8 under the same configuration, consistent with weaker models over-generating causal links at document scale — but with every point a single seed and no pair-level counterpart, we present this section as evidence that SCOTECI *improves with backbone quality*, not as evidence for the specific doc-vs-pair crossover we originally hypothesized. §6.3's cost-effectiveness result is the more solid claim available today.

6.5 Synthetic CoT Interacts with Backbone Reasoning Style

A second, unplanned finding emerged from the CoT-on/off comparisons collected for §6.1: whether synthetic CoT demonstrations help depends on whether the backbone has its own trained reasoning channel. Comparing matched few-shot runs with CoT demonstrations against label-only demonstrations (same dataset, model, and prompt), chat-tuned backbones benefit — `deepseek-v3.2` on Causal-TimeBank: +9.0 F1 (43.1 vs. 34.1, 13 vs. 1 runs); `glm-5.2` on MECI: +16.2 (60.4 vs. 44.2, 1 run each) — while the reasoning-tuned `qwen3-30b-thinking` is consistently *hurt*: -9.7 on MECI (25.6 vs. 35.3, 13 vs. 8 runs) and -6.3 on EventStoryLine (6.0 vs. 12.3, 1 vs. 3 runs). The same reasoning-tuned model also produced five of the seven runs across our logs that collapsed entirely on output format. A plausible mechanism is that models with their own RL-trained reasoning channel are destabilized by in-context reasoning transcripts that were not produced by that channel, while chat models, having no such channel to conflict with, benefit from being shown one. If this holds up under further testing, it argues for routing synthetic-CoT demonstrations by backbone type rather than applying them uniformly, and it directly motivates the teacher-vs-self CoT generation experiment we leave to future work (see Limitations): does a stronger *chat* generator's CoT still help a *reasoning-tuned* consumer, or does the interference come from the transcript itself regardless of its source?

6.6 Error Analysis

Precision rises, not falls, with document density. We originally expected attention dilution: precision degrading as a document accumulates more gold causal pairs. Ranking documents by causal density (gold intra-sentence causal pairs per event mention) into equal-count quartiles and pooling binary-intra precision/recall per bin shows the opposite (Figure 5). Precision climbs monotonically with density on all three datasets — MAVEN-ERE 13.9 (sparsest quartile) to 55.5 (densest), Causal-TimeBank 0.0 to 74.1 — and F1 climbs with it (Causal-TimeBank 0.0 to 53.6; MAVEN-ERE 23.0 to 55.9; EventStoryLine plateaus at 56.3 then 55.2), since the precision gain dominates a mild recall decline. Causal-TimeBank's sparsest quartile — 45 documents with *zero* gold pairs — yields exclusively false positives, and because gold mass concentrates heavily in the dense quartiles (283 of 298 Causal-TimeBank pairs sit in the top two), these sparse-document failures barely dent the pooled headline metric despite being total failures on their own terms. The practical reading is that sparse documents, not dense ones, are SCOTECI's failure mode: the model insists on finding causality where there is little or none, which suggests a cheap mitigation — a sparse-document guard, or restricting unanimous voting (§6.2) to documents below a density threshold — that we leave to future work.

Robustness by backbone. Documents that exhaust all JSON-parsing retries are rare but backbone-dependent: 3.3 per run on `deepseek-v3.2:nitro` (291 over 87 runs) against 0.1 per run on `deepseek-v4-pro` (4 over 38 runs) and 0.6 on the reasoning-tuned `qwen3` variant. The v3.2-to-v4-pro backbone upgrade nearly eliminated parse failures, which also means v3.2-era ablation runs (§6.1) carry a small additional confound, since failed documents score as all- *norel* (§4.5) rather than being excluded.

Causal hallucination is a shared failure mode, not one model's idiosyncrasy. A separate cross-backbone analysis of every intra-sentence prediction logged across the pipeline's candidate backbones (34 qualifying runs spanning 8 models, filtered to total_pairs $>$ 10 and $F1_{\text{intra}} > 0.30$, predominantly on MAVEN-ERE) asks whether different backbones fail independently of one another. They do not: on the 1,294 intra-sentence pairs scored by more than 5 of the 34 runs, predictions agree far above chance (Fleiss' $\kappa = 0.58$ three-way, 0.70 binary), 796 of those 1,294 pairs (61.5%) are wrong in *every* covering run regardless of model, and among the 178 pairs where every covering run unanimously agrees and is wrong, 93% are phantom causality — every model inventing a causal link the annotation says is not there, split almost evenly between the two directions, with essentially no direction confusion. Every one of the 28 cross-model pairs shows positive correctness correlation (ϕ 0.31–0.65), and majority-vote ensembling *across models* recovers only 16.2% accuracy on this hard subset, barely above the single best model (29.3%). This is the mirror image of §6.2's finding: ensembling across demonstration sets works because per-pass false positives are demonstration-specific and largely independent, whereas ensembling across backbones does not fix a bias that all of them already share. It sharpens Gao et al.'s (2023) original causal-hallucination finding: the failure is not one model's idiosyncrasy but a property of the task itself, which is precisely what a document-level, rule-constrained protocol (§4.2) targets directly rather than leaving to backbone choice.

False-positive taxonomy. A keyword-based pass over the false positives of the three headline runs (searching the text between and around each mispredicted pair for causal-marker-adjacent cues) shows that reporting verbs ("said", "according to", "announced") appear near 37.3% of Causal-TimeBank's 150 binary-intra false positives, 30.0% of EventStoryLine's 3,284, and 5.8% of MAVEN-ERE's 5,862; purely temporal connectives ("after", "since", "following") appear near 2.7%, 8.7%, and 10.8% respectively. The transitive-flattening pattern we originally expected to dominate — $A \to B \to C$ read as $A \to C$ — turns out to be minor: only 1.3–2.8% of false positives have a gold $A \to B$ and $B \to C$ edge in the same document (0% on MAVEN-ERE), computed exactly from the gold causal graph rather than by keyword search. The remaining 58–83% of false positives fit neither pattern under this coarse heuristic, so most of SCOTECI's hallucination is not explained by these two surface cues alone and needs a proper manual taxonomy; this keyword pass is a first cut, not a finished analysis.

Qualitative example. On Causal-TimeBank document `APW19980227.0476` (a wire report on a World Court ruling), the model's sentence-by-sentence reasoning correctly rejects several plausible-looking non-causal connections — "that" in *'that has blocked the trial'* is flagged as a relative pronoun, not a causal marker; "to" in *'to force the suspects' extradition'* is flagged as an infinitive of purpose, not an explicit marker — and the pass ends with zero predicted causal pairs. The gold annotation, however, contains exactly the two pairs the model dismissed on those grounds: the objections *caused* the trial to be blocked, and the Security Council resolutions *caused* the push for extradition. Both are genuine causal relations expressed through a purposive or relative-clause construction rather than an explicit connective like "because" or "therefore". This is a concrete instance of the recall cost of the Causal-TimeBank explicit-marker rule (§4.2): a rule strict enough to suppress the reporting-verb and temporal false positives above also suppresses genuine but non-prototypically-worded causal links, one contributor to Causal-TimeBank's recall of 39.6 against its precision of 61.1 (Table 1).

Direction confusion (whether a true positive's predicted direction matches its gold direction) is out of scope under the binary problem definition adopted in §3: since we do not evaluate direction, direction errors are not counted against the metrics we report. For the record, since directed metrics are also logged, predicted direction disagrees with the stored gold direction on 55.1% of EventStoryLine's binary true positives, against 1.9% on MAVEN-ERE and 0.0% on Causal-TimeBank — consistent with EventStoryLine's substantially lower directed (non-collapsed) metrics relative to its binary ones, and a reminder of why we read binary metrics as primary throughout this paper rather than as a simplification of a "truer" directed number.

7 Conclusion

We reframed LLM-based Event Causality Identification from per-pair scaffolding to a single document-level call, guided by a structured reasoning protocol and explicit coherence rules, and introduced synthetic, decontaminated chain-of-thought demonstrations: rationalization used at inference time rather than as fine-tuning data, requiring no human-written rationales, no external knowledge base, and no model training. The result is a large, precision-driven gain where causal hallucination is costliest — +28.1 F1 on Causal-TimeBank, breaking out of the degenerate high-recall/low-precision regime every LLM baseline exhibits there — together with the best precision on MAVEN-ERE and an F1 edge over the strongest prior LLM method on EventStoryLine. Beyond the headline table, treating the method's own logged predictions as data to be analyzed, rather than only as a score to report, surfaced three findings we did not anticipate when designing the method: majority-vote aggregation over demonstration ensembles is empirically F1-optimal rather than merely a safe default; precision *rises*, not falls, with a document's causal density, so sparse documents rather than dense ones are the real failure mode; and synthetic chain-of-thought demonstrations help chat-tuned backbones while measurably hurting a reasoning-tuned one. We take the last of these as evidence that "more reasoning in the prompt" is not a uniformly good design choice, and that scaffold design should account for what kind of reasoning a backbone already does on its own.

The method is, by construction, a bet on model quality: because it asks the model to do exactly the kind of long-context, whole-document reasoning that pair-level scaffolds structurally prevent, we expect it to improve as backbones improve rather than requiring re-engineering around each new model generation. Our current evidence for this (§6.4) is directional and single-seed rather than a controlled crossover result; strengthening it, along with the ablations we cannot yet support (§6.1), is the paper's main unfinished business.

Limitations

Comparability. All SCOTECI numbers in Table 1 are computed on the intra-sentence subset of pairs, matching the subset SERE itself evaluates (§3); this makes the comparison fair rather than incomplete, but it also means full-document (intra- and inter-sentence) performance — a capability the document-level formulation affords precisely because it does not pay per additional pair — is not yet measured for any of the three datasets. Baseline numbers throughout are reproduced from Hao et al. (2026) and Cai et al. (2025) rather than re-run with our backbone, and we do not have a same-backbone pair-level control to separate the document-level formulation's effect from the backbone's; §6.3 explains why that control does not currently exist in our pipeline, and reports the cost-effectiveness argument we use in its place.

Evidence maturity. Several claims in §6.1 and §6.4 rest on single-seed or otherwise confounded runs, calibrated against a measured noise floor of roughly 2–4 F1 on 50-document subsets (§5.4); we have tried to flag every such case rather than present it as settled. Two ablations are currently not just underpowered but structurally unavailable: isolating decontamination requires a configuration option our pipeline does not yet expose (it runs unconditionally in both CoT protocols), and isolating the in-prompt coherence rules requires a controlled comparison we have not yet run. Our exploratory results on MECI, a multilingual ECI corpus the pipeline supports, are noisy enough across repeated identical configurations that we chose not to include them in the main comparison.

Method-specific risks. Decontamination is itself LLM-based, and we have not conducted a manual audit of generated demonstrations to bound residual answer-leakage phrasing that survives it. Synthetic CoT generation costs 2–3 extra LLM calls per unique demonstration document (amortized by caching across a run, but not free), and whether a stronger teacher model can improve the demonstrations consumed by a weaker or differently-tuned reasoner (§6.5) remains untested. Long documents with large pair sets stress the single-call format — output-length limits, and potentially attention dilution on the generation side even though we find no such dilution on the *scoring* side (§6.6) — and per-document routing or chunking may be needed at that extreme, though we have not yet characterized where it starts to bite.

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